

Grade 9 Mathematics - Diagnostic Assessment

Class: IX / SSC-I Sindh | Topics match the Class IX target content PDF

Total: 32 Multiple Choice Questions (2 per unit)

INSTRUCTIONS: Choose the best answer (A, B, C, or D).

UNIT 1: Real and Complex Numbers

1. Which number is irrational?

- A) 0.75
- B) $\frac{2}{7}$
- C) $\sqrt{2}$
- D) 4

2. The conjugate of $3 - 5i$ is:

- A) $3 + 5i$
- B) $-3 - 5i$
- C) $5 - 3i$
- D) $-3 + 5i$

UNIT 2: Logarithms

3. $\log_{10} 1000$ equals:

- A) 2
- B) 3
- C) 10
- D) 100

4. $\log 2 + \log 5$ (common log) equals:

- A) $\log 7$
- B) $\log 3$
- C) 1
- D) 10

UNIT 3: Algebraic Expression and Formulas

5. Which is a polynomial?

- A) $\frac{1}{x} + 2$
- B) $x^2 + 3x + 1$
- C) $\sqrt{x} + 1$
- D) 2^x

6. If $a+b=5$ and $a-b=1$, then ab equals:

- A) 4
- B) 5
- C) 6
- D) 13

UNIT 4: Factorization

7. $x^2 + 7x + 12$ factors as:

- A) $(x+2)(x+6)$
- B) $(x+3)(x+4)$
- C) $(x+1)(x+12)$
- D) $(x-3)(x-4)$

8. Remainder when $x^2 - 3x + 2$ is divided by $x-1$ is:

- A) 0
- B) 1

- C) 2
- D) -1

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UNIT 5: Algebraic Manipulation

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9. HCF of $12x^2y$ and $18xy^3$ is:
- A) $36x^2y^3$
 - B) $6xy$
 - C) $12xy$
 - D) $3x^2y^3$
10. For two expressions, HCF x LCM equals:
- A) HCF + LCM
 - B) the larger expression
 - C) product of the two expressions
 - D) 1

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UNIT 6: Linear Equation and Inequalities

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11. Solve $3x + 7 = 22$.
- A) $x=3$
 - B) $x=5$
 - C) $x=7$
 - D) $x=15$
12. Solving $-2x < 6$ gives:
- A) $x < -3$
 - B) $x > -3$
 - C) $x < 3$
 - D) $x > 3$

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UNIT 7: Linear Graphs and their Applications

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13. The graph of $x = 3$ is:
- A) a horizontal line
 - B) a vertical line
 - C) a line through origin with slope 3
 - D) the x-axis
14. Point $(2, -5)$ lies in quadrant:
- A) I
 - B) II
 - C) III
 - D) IV

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UNIT 8: Quadratic Equations

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15. Solve $x^2 - 9 = 0$.
- A) $x=3$ only
 - B) $x=-3$ only
 - C) $x=3$ or $x=-3$
 - D) $x=9$ or $x=-9$
16. Discriminant of $x^2 + 6x + 9 = 0$ is:
- A) -36
 - B) 0
 - C) 9
 - D) 36

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UNIT 9: Congruent Triangles

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17. Which condition proves congruence of right triangles using hypotenuse and one leg?

- A) SSS
- B) ASA
- C) RHS
- D) AAA

18. AAA for two triangles shows:

- A) congruence always
- B) similarity, not necessarily congruence
- C) equal area only
- D) they are right triangles

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UNIT 10: Parallelograms and Triangles

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19. Diagonals of a parallelogram:

- A) are always equal
- B) bisect each other
- C) are always perpendicular
- D) never intersect

20. Midpoint theorem: the segment joining midpoints of two sides is:

- A) equal to the third side
- B) perpendicular to the third side
- C) parallel to the third side and half as long
- D) twice the third side

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UNIT 11: Line Bisectors and Angles Bisectors

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21. A point on the perpendicular bisector of AB is:

- A) closer to A than to B
- B) equidistant from A and B
- C) the midpoint of AB only
- D) on line AB

22. Angle bisectors of a triangle meet at the:

- A) centroid
- B) orthocentre
- C) circumcentre
- D) incentre

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UNIT 12: Sides and Angles of a Triangle

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23. Which lengths can be sides of a triangle?

- A) 2, 3, 6
- B) 3, 4, 5
- C) 1, 2, 3
- D) 5, 1, 1

24. In a triangle, the side opposite the greatest angle is:

- A) the shortest side
 - B) the longest side
 - C) a median
 - D) always the base
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UNIT 13: Practical Geometry - Triangles

25. SAS construction uses:

- A) three sides
- B) two sides and the included angle
- C) two angles and no side
- D) hypotenuse only

26. Medians of a triangle are concurrent at the:

- A) incentre
- B) circumcentre
- C) centroid
- D) orthocentre

UNIT 14: Theorems Related with Area

27. Area of a triangle with base 10 and height 6 is:

- A) 16
- B) 30
- C) 60
- D) 120

28. Parallelograms on the same base and between the same parallels have:

- A) equal perimeters
- B) equal areas
- C) equal diagonals
- D) congruent shapes

UNIT 15: Projection of a side of a Triangle

29. In a right triangle with legs 3 and 4, the hypotenuse is:

- A) 5
- B) 6
- C) 7
- D) 12

30. Pythagoras $a^2 + b^2 = c^2$ applies when the triangle is:

- A) any triangle
- B) equilateral only
- C) right-angled
- D) obtuse only

UNIT 16: Introduction to Coordinate Geometry / Analytical Geometry

31. Distance between (1,2) and (4,6) is:

- A) 4
- B) 5
- C) 6
- D) 7

32. Midpoint of (2,3) and (6,7) is:

- A) (4,4)
- B) (4,5)
- C) (8,10)
- D) (3,4)

ANSWER KEY

- 1 C $\sqrt{2}$ is irrational
- 2 A conjugate of $a+ib$ is $a-ib$, so $3+5i$
- 3 B $10^3 = 1000$
- 4 C $\log 10 = 1$
- 5 B non-negative integer powers only
- 6 C $4ab = (a+b)^2 - (a-b)^2 = 24$, $ab=6$
- 7 B $3+4=7$ and $3 \times 4=12$
- 8 A $p(1)=0$
- 9 B common lowest powers: $6xy$
- 10 C $\text{HCF} \times \text{LCM} = \text{product}$
- 11 B $3x=15$, $x=5$
- 12 B divide by -2 , reverse inequality
- 13 B $x=\text{constant}$ is vertical
- 14 D $(+,-)$ is quadrant IV
- 15 C $x=\pm 3$
- 16 B $D=36-36=0$
- 17 C RHS
- 18 B AAA is similarity
- 19 B diagonals bisect each other
- 20 C parallel and half
- 21 B equidistant from endpoints
- 22 D incentre
- 23 B $3+4>5$
- 24 B longest side opposite greatest angle
- 25 B two sides and included angle
- 26 C centroid
- 27 B $(1/2) \times 10 \times 6=30$
- 28 B equal areas
- 29 A $9+16=25$
- 30 C right-angled only
- 31 B $\sqrt{9+16}=5$
- 32 B averages $(4,5)$

END OF DIAGNOSTIC